AI-Powered Solar Flare Forecasting: Enhancing Computational Efficiency through the EfficientNet Algorithm

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Abstract **- Solar flares are intense eruptions of electromagnetic radiation and energy resulting from the rapid release of magnetic energy in the solar corona. As a consequence of the emission of high-energy radiation and charged particles, solar flares can disrupt high-altitude aviation operations and satellite communication systems, with the most powerful events occasionally triggering widespread power grid disturbances on Earth. This study presents and evaluates a flare prediction algorithm capable of detecting solar flares up to 24 hours prior to onset, using the EfficientNet architecture and line-of-sight magnetogram data from the Solar Dynamics Observatory's Helioseismic and Magnetic Imager and Geostationary Operational Environmental Satellite X-ray Spectrometer. To produce meaningful training, testing, and validation datasets, the acquired magnetograms were class-balanced through targeted data augmentation and resampling strategies. Fed by individually configured image data generators, binary classification models were trained and evaluated across varying image resolutions, batch sizes, learning rates, and EfficientNet variants. The best-performing EfficientNet algorithm configuration achieved a testing classification accuracy of 85% using only approximately 60,000 training images with an inference rate of around 70 milliseconds per image. Compared to existing approaches, the proposed method demonstrates improved true positive rates while requiring substantially fewer training samples, highlighting the effectiveness of EfficientNet for complex solar image classification tasks and introducing a novel and data-efficient framework for advanced solar flare forecasting.**

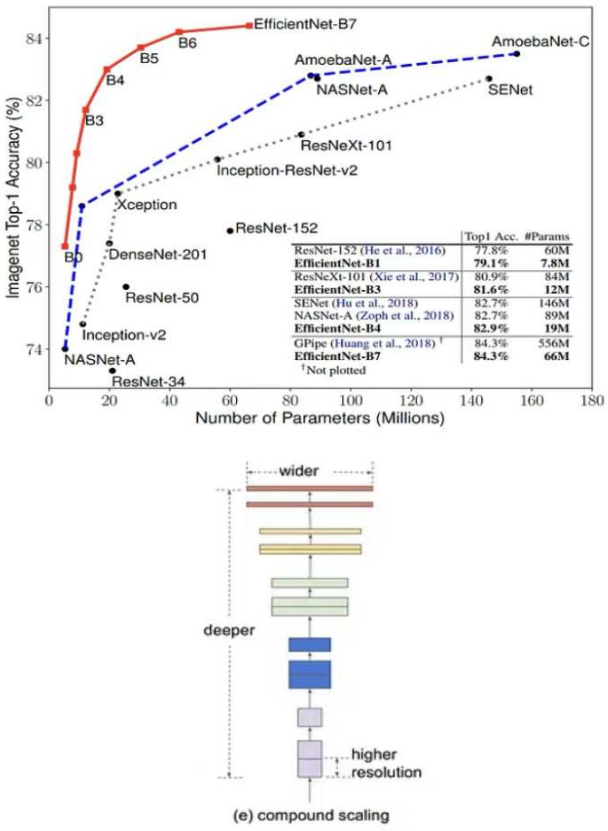
Index Terms - Binary Classification, EfficientNet, Magnetograms, Solar flares

INTRODUCTION

Solar flares occur when the sun's powerful convection currents form coronal loops that burst, releasing energy originally stored in these intense magnetic loops. Flares are classified by the National Aeronautics and Space Administration into A-Class, B-Class, C-Class, M-Class, X-Class by their peak strength of energy released. When directed at Earth, solar activity above C1.0 in magnitude is considered strong enough to cause significant geomagnetic disturbance. Additionally, gamma and x-ray radiation emitted from X-Class and M-Class solar flares can result in solar storms that cause radio blackouts, affect satellites, and endanger astronauts nearby. For instance, in 2017, an X9.3 solar flare that burst amidst Hurricane Irma caused a massive blackout, resulting in devastating consequences for first responders and trapped victims [1]. Because of these potential effects, space researchers and scientists have placed heavy emphasis on the usage of forecasting systems that are able to warn authorities in advance if a flare is imminent and help prevent any associated damages.

The first emergence of forecasting algorithms were physics and statistics based; these methods relied on historical flare trends and produced probabilistic results that often did not generalize well to complex and changing solar magnetic fields [2]. With the advent of machine learning, however, deep learning prediction methods were trained and tested on physical feature datasets, quantitative values calculated from the Solar Dynamics Observatory's magnetogram image data. Although deep learning showed significant improvement over physical models, calculating and processing data values is often computationally time-consuming and frequently requires manual selection; on the other hand, algorithms directly using magnetogram images usually require significantly more data to capture spatial patterns or achieve similar model performance. As a result, developing reliable forecasting systems that are not only accurate, but also optimize efficiency and training time have become increasingly crucial.

Created in 2019 by Mingxing Tan and Quoc V. Le, EfficientNet models are a family of convolutional neural networks (CNNs) designed to achieve higher accuracy while maintaining computational efficiency and requiring fewer model parameters through its compound scaling method [3]. In 2021, EfficientNetV2, which combined the compound scaling method with fused MBConv layers and a progressive learning strategy to further improve training speed and efficiency, was created [4]. Figure I illustrates the superior accuracy and efficiency tradeoff achieved by EfficientNet models compared to other CNN architectures, as well as the compound scaling method that balances network depth, width, and resolution. Leveraging the capabilities of the EfficientNet models with magnetogram image data presents an enhanced approach to solar flare forecasting, aiming to improve predictive accuracy and computational efficiency in data-constrained, real-world applications.



	Top1 Acc.	#Params
ResNet-152 (He et al., 2016)	77.8%	60M
EfficientNet-B1	**79.1%**	**7.8M**
ResNeXt-101 (Xie et al., 2017)	80.9%	84M
EfficientNet-B3	**81.6%**	**12M**
SENet (Hu et al., 2018)	82.7%	146M
NASNet-A (Zoph et al., 2018)	82.7%	89M
EfficientNet-B4	**82.9%**	**19M**
GPipe (Huang et al., 2018) †	84.3%	556M
EfficientNet-B7	**84.3%**	**66M**

†Not plotted

FIGURE I

VARIED PARAMETER EFFICIENTNET MODELS' ACCURACY ON IMAGENET DATA ON THE LEFT AND AN ILLUSTRATION OF EFFICIENTNET'S COMPOUND SCALING ON THE RIGHT [3]

METHODS

Line-of-sight magnetogram images map out both the intensity and direction of the sun's magnetic field visible in the photosphere, with positive and negative polarity of the field represented as with white and black respectively, as shown in Figure II. This study utilizes full resolution line-of-sight solar photospheric magnetogram datasets that were cleaned and processed by [5]. The solar flare image data was initially extracted from NASA's Solar Dynamics Observatory's public database while the binary label classes were generated from the Geostationary Operational Environmental Satellite X-Ray Sensor database. As a result, the positive class consists of magnetogram images with solar flares above the magnitude of C1.0 occurring in the following twenty four hours while the negative class consists of the remaining non-flaring magnetogram images. Overall, the model trained on 59,732 images from the solar active regions of 2284 to 2731. Throughout the study, the image resolutions were varied, with a fixed aspect ratio of 1:1, to determine optimal input configurations.

The full magnetogram dataset was then split into training, testing, and validation subsections for model development and performance evaluation. Aimed at preventing information leakage between the three datasets, solar active regions were assigned exclusively to a single subset, ensuring that no active region appeared in more than one partition.

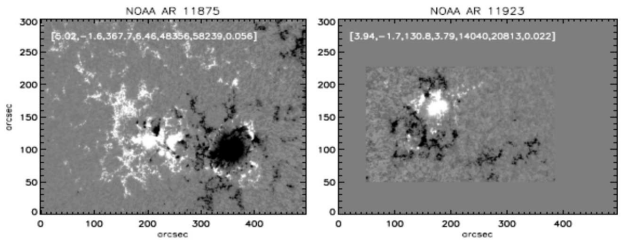


FIGURE II

LINE-OF-SIGHT MAGNETOGRAM IMAGES OF NOAA AR 11875, WHICH PRODUCED 7 C-CLASS FLARES WITHIN 24 HOURS FROM OBSERVATION AND NOAA AR 11923, WHICH PRODUCED NO FLARES [6]

Solar magnetogram data inherently suffers from severe class imbalance, as periods of solar quiescence far outnumber active flaring events. To prevent the model from biasing towards the majority class of non-flaring elements, we implemented a data processing strategy focusing on the minority class of C1.0 or larger flares. This technique simplified and evenly balanced the training and validation set data, with 29,866 elements of each class in total. To better reflect real-world operating conditions, the testing set was intentionally left unbalanced, capturing the intrinsic rarity of strong flares.

An image data generator, which breaks down data into batches to feed into the model, is used on the training, validation, and testing sets. To minimize overfitting, data augmentation techniques, such as random rotation, random flip, and data shuffling, are incorporated into the image data generator between each epoch trained, as visualized in Figure III.

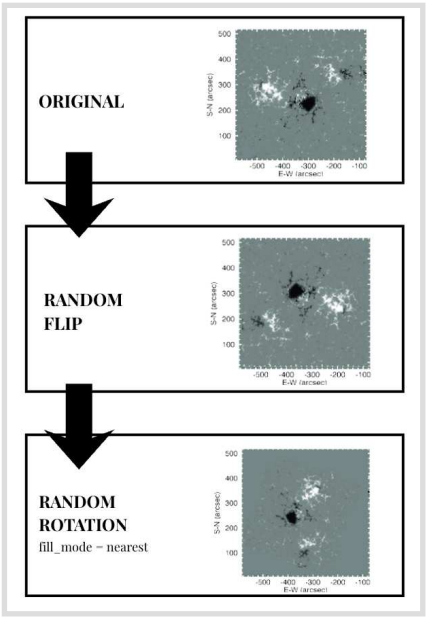


FIGURE III

VARIED PARAMETER EFFICIENTNET MODELS' ACCURACY ON IMAGENET DATA ON THE LEFT AND AN ILLUSTRATION OF EFFICIENTNET'S COMPOUND SCALING ON THE RIGHT [3]

In addition to the random rotation and flipping mentioned, we applied class weighting during the training phase. By assigning a higher penalty to misclassified flare events compared to non-flare events, the EfficientNet algorithm was forced to prioritize feature extraction from active regions. Furthermore, all magnetogram images were normalized to a standard pixel intensity range of 0-1, black and white images, to ensure the model converged based on magnetic field topology rather than luminosity artifacts.

All models were compiled with the Adam optimizer and the Keras framework with the original EfficientNet model heads replaced with a custom classification head consisting of a GlobalAveragePooling2D layer, a 0.5 Dropout layer, and a fully connected softmax output layer. Furthermore, manual early stopping was implemented for models that demonstrated overfitting during training, defined as the epoch when training accuracy began to dip under 0.95 and validation accuracy was less than 0.70. The model training and evaluation workflow is depicted below in Figure IV.

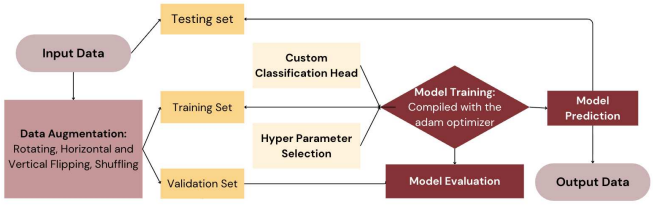


FIGURE IV
FLOWCHART OF THE EFFICIENTNET FLARE FORECASTING MODEL WORKFLOW

In order to optimize model performance, a variety of parameters within the EfficientNet framework were adjusted and tested. For instance, as listed in Table I, these parameters include data resolution, batch size, learning rate, class weights, and EfficientNet model version. To thoroughly assess the performance of each model, this study utilizes evaluation metrics such as Classification Accuracy, F1 score, Sensitivity (True Positive Rate/TPR), Specificity (True Negative Rate/TNR), and Categorical Cross-Entropy Loss. Classification accuracy measures the percentage of correctly predicted instances across all classes, while the F1 score provides a measure of performance in imbalanced datasets. Sensitivity quantifies the percentage of true positive cases correctly identified, whereas Specificity measures the percentage of true negatives correctly classified. TPR and TNR are especially prevalent in flare forecasting tasks to identify whether the model is able to prioritize both minimizing false alarms and maximizing accurate predictions. Finally, Categorical Cross-Entropy Loss evaluates the divergence between the predicted probabilities and the true class labels and serves as the function minimized during model training.

TABLE I
SELECTED MODEL TRAINING CONFIGURATIONS

Model Name	EfficientNet Version	Learning Rate	Batch Size	Image Resolution	Class Weights
M1	B7	0.0001	32	224	1:1
M2	B7	0.001	16	256	1:1
M3	B7	0.001	16	260	2:1
M4	V2B2	0.001	16	260	1:1
M5	V2B2	0.001	16	260	1:5
M6	V2B3	0.001	16	300	1:1.5

RESULTS AND DISCUSSION

All EfficientNet models performed increasingly well on the training data, with the EfficientNetV2 models reaching training loss values of 0.1 and EfficientNetB7 models reaching training loss values of around 0.5 within five epochs. The EfficientNetB7 models performed slightly better on the validation set, with the highest validation classification accuracy as 83% in M1. On the other hand, EfficientNetV2 models achieved slightly lower validation classification accuracies of 81%. Evident in both validation and testing and as illustrated in Table II and Figure V and VI, EfficientNetB7 usually showed a high TPR and low TNR while EfficientNetV2 models usually showed a higher TNR and lower TPR.

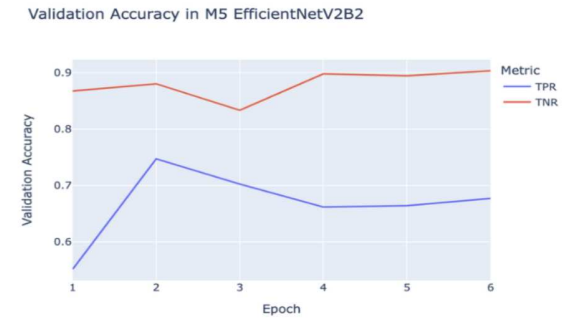


FIGURE V
VALIDATION SET TPR AND TNR PROGRESSION IN THE M5 EFFICIENTNETV2B2 MODEL ACROSS SIX EPOCHS

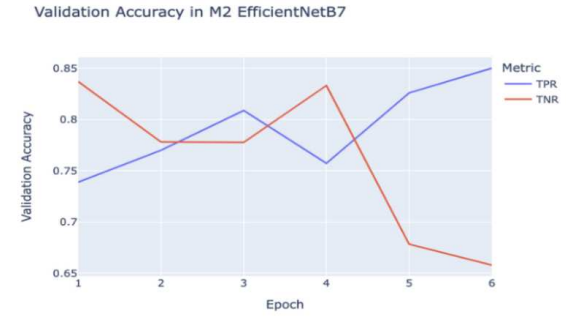


FIGURE VI
VALIDATION SET TPR AND TNR PROGRESSION IN IN THE M2 EFFICIENTNETB7 MODEL ACROSS SIX EPOCHS

TABLE II
TESTING RESULTS FOR VARIED MODEL CONFIGURATIONS

Model Name	True Positive Rate	True Negative Rate	Overall Classification Accuracy
M1	0.8909	0.5819	0.6880
M2	0.7838	0.8561	0.8486
M3	0.6794	0.8621	0.8194
M4	0.5535	0.9128	0.8164
M5	0.6876	0.8648	0.8235
M6	0.5765	0.9097	0.8206

Compared to previous research, and as graphed in Figure VII and Figure VIII, EfficientNetB7 shows notable promise and potential with the best performing M2 model achieving similar TPR to other novel approaches and even exceeding the ResNet-18, SqueezeNet, and VGG-16 models' TPR with only an eighth of the data in some cases [7]-. Along with this, the study performed by Boucheron and others applies a baseline VGG-16 model to their full resolution line-of-sight magnetogram dataset, the same dataset used in this study, and reports a TPR of 69.52% and TNR of 81.42%, providing additional insights into how the M2 EfficientNet model delivers improved performance over VGG-16 models [5]. In particular, Figure IX reveals how M2 achieves high predictive accuracy, correctly identifying 15,022 non-flare cases and 4,573 flare cases on the unbalanced testing dataset, while misclassifying only 2,525 non-flare cases as flares and 1,261 flare cases as non-flares.



FIGURE VII
COMPARISON OF TOTAL DATA VOLUME UTILIZED ACROSS STUDIES [7-10]

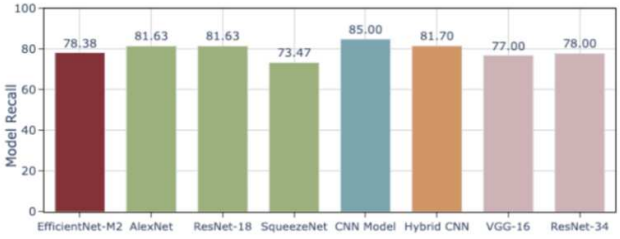


FIGURE VIII
MODEL RECALL PERFORMANCE ACROSS DIFFERENT ARCHITECTURES [7-10]

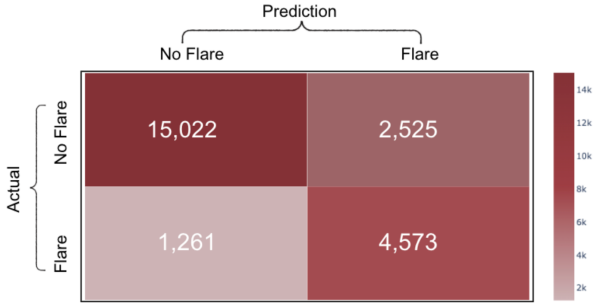


FIGURE IX
CONFUSION MATRIX OF THE BEST-PERFORMING M2 EFFICIENTNETB7 MODEL'S TESTING SET PERFORMANCE

Furthermore, the M2 model averaged training times of ~200 ms per image and inference times of ~70 milliseconds per image on easily affordable and widely used TPU systems, making them especially useful for forecasting tasks that value efficiency and real-time forecasts.

While previous studies have utilized architectures such as VGG-16 and ResNet-50 for solar weather forecasting, these models often suffer from parameter bloating. For instance, VGG-16 contains approximately 138 million parameters. In contrast, our implemented EfficientNet-B3 utilizes compound scaling to achieve superior accuracy with significantly fewer parameters, approximately 12 million. This reduction is critical for the real-time application of this algorithm in space weather forecasting, as a lightweight model allows for potential faster deployment. By maintaining an 85% accuracy while reducing parameter count by over 90% compared to previous comparable architectures, this study demonstrates that data-efficient deep learning can be both highly accurate and computationally viable for real-time space weather monitoring and flare forecasting.

CONCLUSION

The findings of this study suggest that EfficientNetV2 models, particularly EfficientNetV2B2 and EfficientNetV2B3, exhibit better True Negative Rate compared to True Positive Rate and achieve optimal performance when trained with class weights favoring the positive class and larger image resolutions. On the other hand, the EfficientNetB7 models tended to achieve higher TPR than TNR and performed best when trained with balanced class weights, smaller batch sizes, and larger image resolutions.

Furthermore, of all EfficientNet models trained, the M2 EfficientNetB7 shows the most balanced and promising performance on both TPR and TNR, reaching an overall classification accuracy of 85%. Training with a smaller learning rate for EfficientNetB7 corresponded to higher validation accuracy and high testing set TPR, indicating how smaller learning rates improve generalization accuracy but simultaneously increase training time. Similarly, decreasing batch size in EfficientNetV2B2 improved validation classification accuracy by a maximum of 4%, as it allowed the model to easily capture nuances in the data by inputting

smaller quantities of data at a time. EfficientNetB7 models were more sensitive to unbalanced class weights and fluctuated more between having higher TPR or TNR, while EfficientNetV2B2 was less sensitive and demonstrated more consistent performance across varying class weight configurations.

Overall, in comparison to other solar flare prediction techniques, the EfficientNet model demonstrates substantial potential in large-scale image classification tasks, particularly due to its ability to prioritize either detecting solar flares or minimizing false alarms while achieving high accuracies. Furthermore, by reducing the reliance on large training datasets, EfficientNet models can significantly enhance the speed and cost-effectiveness of forecasting efforts, ultimately improving our ability to mitigate the risks associated with space weather hazards. In practical terms, this advancement could lead to more timely and accurate alerts for solar flare events, which pose significant threats to satellite operations, GPS navigation systems, aviation, and power grids on Earth. With improved forecasting capabilities powered by EfficientNet, space weather monitoring systems can become more proactive, allowing stakeholders, such as utility companies, aerospace agencies, and emergency responders, to implement mitigation strategies before disruptions occur. Furthermore, the model's scalability and efficiency suggest that it could be adapted for other geophysical hazard prediction tasks, enhancing resilience in various fields and broader applications.

Potential future advancements include extending EfficientNet-based flare forecasting systems to multi-class classification frameworks that explicitly account for the varying intensity magnitudes of solar flares. In addition, incorporating physical and magnetic parameters derived from Solar HMI Active Region Patches show potential in providing richer and more physically meaningful input representations for model training. To further capitalize on the complementary strengths of the range of EfficientNet architectures evaluated in this study, future work may explore the development of hybrid modeling frameworks that integrate multiple EfficientNet variants within a unified architecture.

Acknowledgment

I would like to acknowledge the use of open-source resources, including Python libraries such as MatPlotLib, Tensorflow, Numpy, Pandas, and AstroPy, which supported my independent work on this project. Additionally, I would like to thank the Boucheron et al. (2023) study for providing the raw data that contributed to this research.

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